

AMAZING MIDDLE SECTION

SAQ

SECTION-C

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SAQ CHAPTERS

28 PAGES - 32 MARKS

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Super Scoring Pages

★★★SSP★★★

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DEAR DUDE!

Please note that SAQs in each chapter are Jumbled, and Easy SAQs are brought to the beginning.

1 SOLUTIONS

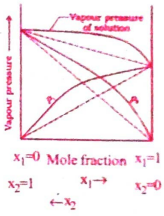
4 Marks

12. What is meant by positive deviations from Raoult's law and how is the sign of Δ_{mix} H related to positive deviation from Raoult's law ? [BMP]

A: 1) Positive deviation occurs when the vapour pressure of a solution is higher than the predicted value by Raoult's law.

2) In such cases, intermolecular interactions between solute and solvent particles (A and B) are weaker than those between solute-solute (A-A) and solvent-solvent (B-B).

3) It leads to increase in vapour pressure resulting in positive deviation.



4) For a solution showing positive deviation

(i)$P_{total} > P_A+P_B$ (ii)$\Delta H_{mix} > 0$ (+Ve) (iii)$\Delta V_{mix} > 0$ (+Ve)

5) **Ex:** (i) Ethyl alcohol and water (ii) Acetone and benzene.

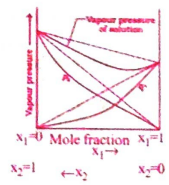
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13. What is meant by negative deviation from Raoult's law and how is the sign of Δ_{mix} H related to negative deviation from Raoult's law ?

A: 1) Negative deviation occurs when the vapour pressure of a solution is lower than the predicted value by Raoult's law.

2) In such cases, intermolecular interactions between solute and solvent particles (A and B) are stronger than those between solute-solute (A-A) and solvent-solvent (B-B).

3) It leads to decrease in vapour pressure resulting in negative deviation.



4) For a solution showing negative deviation

(i) $P_{total} < P_A + P_B$; (ii)$\Delta H_{mix} < 0$, (–Ve) (iii)$\Delta V_{mix} < 0$, (–Ve)

5) **Ex:** (i) Nitric acid and water (ii) Hydrochloric acid and water.

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14. What is relative lowering of vapour pressure? How is it useful to determine the molar mass of a solute?

A: **1) Lowering of Vapour Pressure(LVP):** It is the difference between the vapour pressure of pure solvent (p^o) and the vapour pressure of the solution (p^s).

Thus, $\Delta p = p^o - p^s$

2) Relative Lowering of Vapour Pressure(RLVP): It is the ratio between lowering of vapour pressure ($p^o - p^s$) and vapour pressure of pure solvent (p^o).

Thus, R.L.V.P $= \frac{p^o - p^s}{p^o}$

3) According to Raoult's law, RLVP = mole fraction of the solute (χ_s.)

$$\frac{p^o - p^s}{p^o} = \chi_s \Rightarrow \frac{p^o - p^s}{p^o} = \frac{w_s}{M_s} \times \frac{M_o}{w_o}$$

4) From the above relation, molar mass of the given solute is $M_s = \frac{w_s \times M_o \times p^o}{(p^o - p^s) w_o}$

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15. Calculate the mass of a non-volatile solute (molar mass 40g mol^{-1}) which should be dissolved in 114g Octane to reduce its vapour pressure to 80%. [AP 20]

A: Vapour pressure is reduced 80% means vapour pressure of the solution reduces to 80.

Vapour pressure of solvent P^o=100, Vapour pressure of solution P^s=80

Mass of solvent (octane) (w_o)=114g, Molar mass of Octane (C_8H_{18}) is M_o=114 g/mol

Molar mass of solute is M_s=40 g/mol, Mass of solute is w_s =?

From Raoult's law, $\frac{P^o - P^s}{P^o} = \frac{w_s}{M_s} \times \frac{M_o}{w_o}$

$$\Rightarrow \frac{100-80}{100} = \frac{w_s}{40} \times \frac{114}{114} \Rightarrow \frac{20}{100} = \frac{w_s}{40} \Rightarrow w_s = \frac{20 \times 40}{100} = 8\ g$$



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16. Define mass percentage, volume percentage and mass to volume percentage solutions.

A: **(i) Mass percentage (w/w):** Mass of the component (in g) present in 100g of solution.

$$\text{Mass \% of a component} = \frac{\text{Mass of the component in the solution}}{\text{Total mass of the solution}} \times 100$$

(ii) Volume percentage (V/V): Volume of the component (in mL) present in 100 mL of solution.

$$\text{Volume \% (V/V) of a component} = \frac{\text{Volume of the component in the solution}}{\text{Total volume of the solution}} \times 100$$

(iii) Mass by volume % (w/V): The mass of solute in (g) present in 100 mL of solution.

$$\text{Mass by volume \% (w/V) of solute} = \frac{\text{Mass of the solute in the solution}}{\text{Total volume of the solution}} \times 100$$

---

17. A solution of sucrose in water is labeled as 20% w/w. What would be the mole fraction of each component in the solution ?

A: 20% w/w sucrose solution means 20g of sucrose is present in 100g of solution.
Weight of sucrose solute = 20g, GMM of sucrose ($C_{12}H_{22}O_{11}$)= 342 g mol^{-1}.

Moles of sucrose, $n_A = \frac{\text{weight}}{\text{GMM}} = \frac{20}{342} = 0.0585$

Weight of solution = 100g, Mass of solvent = 100–20 = 80g,
GMM of water (H_2O) = 18 g mol^{-1}.

Moles of water, $n_B = \frac{\text{weight}}{\text{GMM}} = \frac{80}{18} = 4.4445$

Mole fraction of sucrose $(\chi_A) = \frac{n_A}{n_A + n_B} = \frac{0.0585}{0.0585 + 4.4445} = \frac{0.0585}{4.503} = 0.013$

Mole fraction of water $(\chi_B) = 1 - \chi_A = 1 - 0.013 = 0.987$ $\left[\because \chi_A + \chi_B = 1\right]$

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18. A solution of glucose in water is labeled as 10% w/w. What would be the molality of the solution ? [AP 20,25]

A: 10% w/w glucose solution means 10g of glucose is dissolved in 100g of solution.

Weight of Glucose solute (w_s)=10 g, GMM of glucose =180,

Moles of glucose $n = \frac{\text{weight}}{\text{GMM}} = \frac{10}{180}$, Weight of solvent (W)= 100–10=90g

Molality of the solution $m = n \times \frac{1000}{W\ (g)} = \frac{10}{180} \times \frac{1000}{90} = 0.617m$

∴ Molality of the given solution =0.617m

[☞ Remaining SAQs are in Page 87]

2 ELECTRO CHEMISTRY

4+4 Marks

19. What are Galvanic cells? Explain the working of Galvanic cell with a neat sketch taking Daniel cell as example. **[BMP][AP 15,18]**

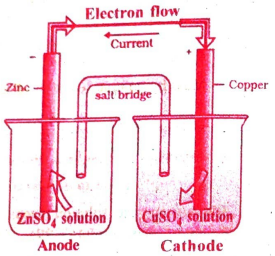
A: **1) Galvanic Cell:** It is an electro chemical cell which converts chemical energy into electrical energy by redox reactions (Spontaneous).

2) Construction: (i) Daniel Cell consists of two half cells (a)anode half cell (b)cathode half cell.

(ii) The two half cells are connected by a salt bridge containing a saturated solution of KNO_3 in agar-agar gel.

(iii) The anode half cell consists of zinc rod dipped in $ZnSO_4$ solution.

(iv) The cathode half cell consists of copper rod dipped in $CuSO_4$ solution.



DANIEL CELL

3) Representation: $Zn_{(s)} / Zn^{+2}_{(aq)} // Cu^{+2}_{(aq)}/Cu_{(s)}$

4) Working: When the two half cells are connected externally through a voltmeter, current flows out from the cell due to potential difference.

5) Reactions:

Oxidation reaction at anode : $Zn \rightleftharpoons Zn^{+2} + 2e^-$

Reduction reaction at cathode: $Cu^{+2} + 2e^- \rightleftharpoons Cu$

6) Net cell reaction: $Zn_{(s)} + Cu^{+2}_{(aq)} \rightleftharpoons Zn^{+2}_{(aq)} + Cu_{(s)}$

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20. State and explain Kohlrausch's law of independent migration of ions. **[BMP] [AP 15,16,19,22,23,25]**

A: **1) Kohlrausch law of independent migration of ions** states that " limiting molar conductivity of an electrolyte can be represented as the sum of the individual contributions of the anion and the cation of the electrolyte, at infinite dilution."

2) Thus, if $\lambda^0_{Na^+}$ and $\lambda^0_{Cl^-}$ are the limiting molar conductivities of the sodium and chloride ions respectively, then the limiting molar conductivity for sodium chloride is given by the equation

$$\Lambda^\circ_{(NaCl)} = \lambda^0_{Na^+} + \lambda^0_{Cl^-}$$

3) In general, if an electrolyte on dissociation gives V_+ cations and V_- anion then its limiting molar conductivity is given by $\Lambda^\circ_m = V_+\lambda^0_+ + V_-\lambda^0_-$

4) Here, λ^0_+ and λ^0_- are the limiting molar conductivities of the cation and anion respectively.

21. Give the construction and working of a standard hydrogen electrode with a neat diagram.

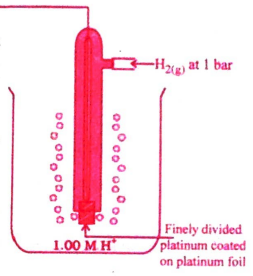
A: **Standard hydrogen electrode (SHE) :**

1) SHE is a reference electrode and its reduction potential is arbitrarily assigned as zero volt at all temperatures.

2) By using 'SHE' we can measure the potential of single electrode.

3) H_2 gas adsorbed on platinum at one atmosphere placed in contact with 1M H^+ concentration in solution is treated as standard hydrogen electrode (SHE).

4) Potential of this electrode is taken as zero.



Hydrogen electrode is represented as $Pt, H_{2(g)}\ 1\ atm / H^+_{(aq)} (C = 1)$

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22. What are the products obtained at the cathode and the anode during the electrolysis of the following when platinum electrodes are used in the electrolysis

(a) Molten KCl (b) Aq. $CuSO_4$ solution (c) Aq.K_2SO_4 solution

A: **a) Electrolyte:** $2KCl \xrightarrow[\text{Molten}]{} 2K^+ + 2Cl^-$

at cathode: $2K^+ + 2e^- \rightarrow \mathbf{2K}$

at anode : $2Cl^- \rightarrow \mathbf{Cl_2} + 2e^-$

b) Electrolyte: $CuSO_{4(aq)} \rightarrow Cu^{+2}_{aq} + SO^{-2}_{4(aq)}$

at cathode: $Cu^{+2} + 2e^- \rightarrow \mathbf{Cu(s)}$

at anode: $H_2O \rightarrow 2H^+ + \frac{1}{2}\mathbf{O_2} + 2e^-$

c) Electrolyte: $K_2SO_{4(aq)} \rightarrow 2K^+_{aq} + SO^{-2}_{4(aq)}$

at cathode: $2H_2O + 2e^- \rightarrow 2OH^- + \mathbf{H_2}$

at anode: $H_2O \rightarrow 2H^+ + \frac{1}{2}\mathbf{O_2} + 2e^-$

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23. State and explain Nernst equation with the help of a metallic electrode and a non-metallic electrode.

A: Nernst Equation: The equation that gives the dependence of the electrode potential on the concentration of ions with which electrode is reversible is known as Nernst equation.

Nernst Equation: Electrode potential $E = E^\circ - \frac{2.303RT}{nF}\log\frac{[Products]}{[Reactants]}$

(a) For Metal Electrode: Electrode reaction: $M^{n+}_{(aq)} + ne^- \rightleftharpoons M_{(s)}$

$$E = E^\circ - \frac{2.303RT}{nF}\log\frac{1}{[M^{n+}]} = E^\circ + \frac{2.303RT}{nF}\log[M^{n+}]$$

At 25°C, $\frac{2.303RT}{F} = \frac{2.303\times 8.314\times 298}{96500} = 0.059$

$\therefore E = E^\circ + \frac{0.059}{n}\log C$ [Here C= $[M^{n+}]$ = Metal ion concentration]

(b) For Non -metal Electrode: Electrode reaction: $\frac{1}{2}A_{2(g)} + ne^- \rightleftharpoons A^{-n}_{(aq)}$

$E = E^\circ - \frac{2.303RT}{nF}\log[A^{-n}]$

At 25°C, $E = E^\circ - \frac{0.059}{n}\log C$ [Here C = $[A^{-n}]$ Anion concentration]

Notation: E°= Standard reduction potential of the cell at 25°C.

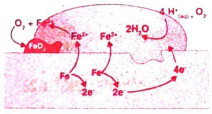
n= Number of electrons involved in cell reaction. R= Gas constant R (8.314 J K^{-1} mol^{-1});

T= Temperature =273 + 25= 298K; F = Faraday constant= 96,487 = 96500 c mol^{-1};

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24. What is metallic corrosion? Explain it with respect to iron corrosion.

A: 1) Metallic corrosion: It is the process of slow eating up of metals by gases and water vapours present in atmosphere due to the formation of certain compounds like oxides, sulphides, carbonates,etc.,



2) Corrosion of Iron is known as Rusting. Rusting is an electrochemical process that occurs when iron is exposed to oxygen and moisture.

3) Electrochemical Mechanism:

i) Formation of Anodic and Cathodic Regions

ii) On the iron surface, tiny regions act as:Anode→ oxidation occurs;Cathode → reduction occurs

4) The corrosion process is electrochemical in nature. On the iron surface, iron becomes oxidized in the presence of water.

5) On the metal surface, oxidation takes places at anode and reduction takes place at cathode.

At anode: $Fe_{(s)} \longrightarrow Fe^{+2}_{aq} + 2e^-$ (Iron atoms lose electrons and go into solution)

At cathode: $\frac{1}{2}O_{2(aq)} + H_2O_{(l)} + 2e^- \longrightarrow 2OH^-_{aq}$ (Oxygen reacts with water to form hydroxide ions)

Overall reaction: $Fe_{(s)} + \frac{1}{2}O_{2(aq)} + H_2O_{(l)} \longrightarrow Fe^{+2}_{aq} + 2OH^-_{aq}$

6) Formation of Rust: Ferrous ions react with hydroxide ions $Fe^{+2}_{aq} + 2OH^- \longrightarrow Fe(OH)_2$

On further oxidation $Fe(OH)_2 \longrightarrow Fe(OH)_3$

The Ferrous ions finally form hydrated ferric oxide $Fe_2O_3.xH_2O$ (rust).

PROBLEMS

25. A solution of $CuSO_4$ is electrolysed for 10 minutes with a current of 1.5 amperes. What is the mass of copper deposited at the cathode? [AP 15,18,20]

Sol: Given time t = 10 min. = 600s, current i= 1.5 amperes

charge = current × time =1.5 × 600 = 900C

According to the reaction: $Cu^{2+}(aq) + 2e^- = Cu(s)$

We require 2F or 2×96500 C deposit 1 mole or 63 g of Cu.

For 900C, the mass of Cu deposited = (63 ×900)/(2×96500) = 0.2938 g

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26. Calculate Λ^0_m for $CaCl_2$ and $MgSO_4$ from the given data Ca^{2+} = 119.0S cm^2 mol^{-1}. Cl^- = 76.3S cm^2 mol^{-1} Mg^{2+} = 106.0 S cm^2 mol^{-1} SO_4^{2-} =160.0S cm^2 mol^{-1}

Sol: We know from Kohlrausch law that

$\Lambda^0_{m(CaCl_2)} = \lambda^0_{Ca^{2+}} + 2\lambda^0_{Cl^-} = 119.0 + 2(76.3) = 119.0 + 152.6 = 271.6$ S cm^2 mol^{-1}.

$\Lambda^0_{m(MgSO_4)} = \lambda^0_{Mg^{2+}} + \lambda^0_{SO_4^{2-}} = 106.0 + 160.0 = 266$ S cm^2 mol^{-1}.

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27. The standard emf of Daniel cell is 1.1V. Calculate the standard Gibbs energy for the cell reaction: [AP 15]

$Zn(s) + Cu^{2+}(aq) \rightarrow Zn^{2+}(aq) + Cu(s)$

Sol: $Zn(s) + Cu^{2+}(aq) \rightarrow Zn^{2+}(aq) + Cu(s)$

From above equation n=2, F=96500C, E°_{Cell} = 1.1 V.

$\Delta G^\circ_F = -nFE^\circ_{Cell}$

$\Delta G^\circ_F = -2\times 96500\times 1.1 = -21200 J.mole^{-1} = -212.3$ KJ.$mole^{-1}$



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[☞ Remaining SAQs are in Page 88]

3 CHEMICAL KINETICS

4 Marks

28. What is 'molecularity' of a reaction? How is it different from the 'order' of a reaction? Name one bimolecular and one trimolecular gaseous reactions. [BMP][AP 19,20,23]

A: **1) Molecularity** is the number of reacting molecules taking part in a single elementary step.

2) Order of a reaction is the 'sum of powers of the concentration terms' of the reactants in the rate law equation.

3) Differences between Molecularity and Order :

Molecularity	Order
i) It is determined **theoretically**.	i) It is determined **experimentally**.
ii) It can have **integral values** only.	ii) It can have **fractional values** also.
iii) It can be unimolecular, bimolecular....	iii) It can be zero order, first order....
iv) It is applicable only to **elementary reactions**.	iv) It is applicable to **elementary and complex reactions.**

4) Bimolecular reaction: Dissociation of hydrogen iodide into H_2 and I_2.

$2HI \rightarrow H_2 + I_2$

5) Trimolecular reaction: Formation of NO_2 from NO and O_2

$2NO + O_2 \rightarrow 2NO_2$

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29. Derive the integrated rate equation for a zero order reaction.

A: 1) Consider a zero order reaction : $R \rightarrow$ product

$\therefore$ Rate $= \frac{-d[R]}{dt} = k[R]^0 = k \Rightarrow d[R] = -kdt$

2) Integrating both sides, we get $[R] = -kt + I$.....(1), I is the constant of integration

At $t = 0$, we have $[R] = [R]_0$, where $[R]_0$ is initial concentration of the reactant.

3) Equation (1) can be written as $[R]_0 = -k \times 0 + I \Rightarrow [R]_0 = I$

Substituting the value of I in the equation (1), we have $[R] = -kt + [R]_0$

[This is in the form y=mx+C of a line, with slope m= –k]

4) $\Rightarrow kt = [R]_0 - [R] \Rightarrow k = \frac{[R]_0 - [R]}{t}$. This is the integrated equation for zero order reaction.

Here, k= rate constant ; $[R]_0$ = Initial concentration of the reactant

t= time, [R]= Final concentration of reactants at the time t.

30. Derive an integrated rate equation for a first order reaction. [AP 22]

A: 1) Consider a first order reaction: $R \rightarrow$ Product

$\text{Rate} = \frac{-d[R]}{dt} = k[R] \Rightarrow \frac{d[R]}{[R]} = -kdt$

2) Integrating this equation, we get $\ln[R] = -kt + I$.....(1), where I is the constant of integration

When $t = 0$ we have $[R] = [R]_0$, where $[R]_0$ is the initial concentration of the reactant.

3) Equation (1) can be written as $\ln[R]_0 = -k \times 0 + I \Rightarrow \ln[R]_0 = I$

Substituting the value of I in equation (1), we have

4) $\ln[R] = -kt + \ln[R]_0 \Rightarrow kt = \ln[R]_0 - \ln[R] = \ln\frac{[R]_0}{[R]} \Rightarrow k = \frac{1}{t}\ln\frac{[R]_0}{[R]} \Rightarrow k = \frac{2.303}{t}\log\frac{[R]_0}{[R]}$

This is the integrated rate equation for first order reaction.

Here, k= rate constant ; $[R]_0$ = Initial concentration of the reactant

t= time, [R]= Final concentration of the time t.

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31. What is half-life ($t_{1/2}$) of a reaction? Derive the equations for the 'half-life' value of zero and first order reactions.

A: **Half -life of a reaction ($t_{1/2}$)** is the time in which the concentration of a reactant is reduced to one half of its initial concentration.

(i) For a zero order reaction $k = \frac{[R]_0 - [R]}{t}$

At $t = t_{1/2}$ we have $[R] = \frac{[R]_0}{2}$

$\therefore$ Rate constant is $k = \frac{[R]_0 - \frac{1}{2}[R]_0}{t_{1/2}} \Rightarrow k = \frac{\frac{[R]_0}{2}}{t_{1/2}} \Rightarrow t_{1/2} = \frac{[R]_0}{2k}$

(ii) For a first order reaction $k = \frac{2.303}{t}\log\frac{[R]_0}{[R]}$

At $t_{1/2}$ we have, $[R] = \frac{[R]_0}{2}$

$\therefore$ Rate constant is $k = \frac{2.303}{t_{1/2}}\log\frac{[R]_0}{[R]_0/2} \Rightarrow k = \frac{2.303}{t_{1/2}}\log 2$

$\Rightarrow t_{1/2} = \frac{2.303}{k}\log 2 = \frac{2.303}{k} \times 0.3010 = \frac{0.693}{k}$ $\quad \therefore t_{1/2} = \frac{0.693}{k}$

32. Explain the terms

a) Activation energy (E_a) **b)Collision frequency (Z)**

c) Probability factor (P) with respect to Arrhenius equation.

A: **a) Activation Energy:** It is the minimum excess energy that reactant molecules should acquire to participate in chemical reaction or to cross threshold energy barrier. $Ea_a = E_T - E_R$.

In other words, activation energy is the difference between threshold energy and the average kinetic energy of the reactant molecules.

b) Collision Frequency is the number of collisions that take place per second per unit volume of the reaction mixture.

c) Probability Factor: For a collision to be effective, the colliding molecules must have energy greater than threshold energy and also must have proper orientations.

For the collision to be effective, another factor, P is called probability factor has to be introduced.

Thus Arrhenius equation is modified to the form $k = PZ_{AB}\, e^{-Ea/RT}$

where P= Probability factor

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33. What is Arrhenius equation? Derive an equation which describes the effect of rise of temperature (T) on the rate constant (k) of a reaction.

A: **1) Arrhenius equation** gives the temperature dependence of the rate of a chemical reaction.

Arrhenius equation: Rate constant $k = A\, e^{-Ea/RT}$.....(1)

Where A is frequency factor, E_a activation energy.

2) Taking natural logarithm on both sides of equation (1), we get $\ln k = \frac{-E_a}{RT} + \ln A$ .....(2)

3) At temperature T_1, equation (2) is $\ln k_1 = \frac{-E_a}{RT_1} + \ln A$ .....(3)

4) At temperature T_2, equation (2) is $\ln k_2 = \frac{-E_a}{RT_2} + \ln A$ .....(4)

Here K_1 and K_2 are the values of rate constants at temperature T_1 and T_2 respectively.

5) Subtracting equation (3) from (4), we get $\ln k_2 - \ln k_1 = \frac{E_a}{RT1} - \frac{E_a}{RT_2}$

$$\ln\frac{k_2}{k_1} = \frac{E_a}{R}\left[\frac{1}{T_1} - \frac{1}{T_2}\right] \Rightarrow 2.303\log\frac{k_2}{k_1} = \frac{E_a}{R}\left[\frac{1}{T_1} - \frac{1}{T_2}\right] \Rightarrow \log\frac{k_2}{k_1} = \frac{E_a}{2.303R}\left[\frac{1}{T_1} - \frac{1}{T_2}\right]$$

4 The d&f-Block Elements

4 Marks

34. Write the characteristics properties of transition elements. [BMP] [AP 15]

A: **The characteristics properties of Transition metals:**

1) Transition metals are 'hard and heavy' metals.
2) They show 'variable oxidation' states.
3) They are 'good conductors of heat and electricity'.
4) They exhibit 'paramagnetic character'.
5) They form 'alloys'.
6) They form 'coloured complex compounds'.

☺ TRANSITION PROPERTIES ☺
H
O
T
P
A
C

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35. What is lanthanoid contraction? What are the consequences of lanthanoid contraction?

A: **Lanthanoid contraction:** It is the steady decrease in the atomic radii even with increase in atomic number from Lanthanum to Lutetium due to **poor screening effect of 4f–electrons**.

Consequences of Lanthanoid contraction:

1) **Basic character of oxides and hydroxides:** Due to the lanthanoid contraction, the covalent nature of La-OH bond increases and thus, the basic character of oxides and hydroxides decreases from $La(OH)_3$ to $Lu(OH)_3$.
2) Due to lanthanide contraction, size of 4d series and 5d series elements remains the same. Thus, pairs of elements such as Zr/Hf, Nb/Ta and Mo/W are almost identical in size. Due to almost similar size, such pairs have very similar properties which makes their separation difficult.
3) **Separation of lanthanoids:** Due to lanthanoid contraction, there is a difference in some properties of lanthanoid like solubility, degree of hydration and complex formation. These differences enable the separation of lanthanoids by 'ion exchange method'.

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36. What are interstitial compounds? How are they formed? Give two examples. [AP 15]

A: **1) Interstitial compounds** are compounds formed when small atoms (H,C,B,N) occupy the empty spaces (interstitial sites-voids) in the metals of crystal lattice structure.

2) Formation: In metals such as Fe, Ti atoms are arranged in a closely packed structures with tiny gaps (voids). Small atoms can fit in this voids without disturbing the overall structure much.

Ex 1: Fe-C (Iron- carbon steel),

Ex 2: Ti-H (Titanium - Hydride), Fe_3H

37. Write the characteristics of interstitial compounds.

A: Characteristics of interstitial compounds:

1) They are very hard and rigid. **Ex:** Some borides approach diamond in hardness.
2) They have high melting points, higher than those of their pure metals.
3) They retain metallic conductivity
4) They are chemically inert.

---

38. Why do the transition metal ions exhibit characteristic colours in aqueous solution. Explain giving examples.

A: 1) Transition metal ions exhibit **colour property** in aqueous solution due to the **presence of unpaired d-electrons.**

2) These unpaired d-electrons from a **lower energy** are excited to a higher energy d-orbitals.
3) The energy of excitation corresponds to the frequency of light absorbed in visible region.
4) The colour observed corresponds to the complementary colour of light absorbed.
5) The frequency of light absorbed is determined by the nature of the ligand.

Ex1: Ti^{2+} [Ar] $3d^1$ Purple

V^{2+} [Ar] $3d^3$ Violet

These have **unpaired electrons** in their d orbitals and hence they are **coloured.**

Ex2: Sc^{3+} [Ar] $3d^0$ colourless

Cu^{+} [Ar] $3d^{10}$ colourless

These have **paired electrons** in their d orbitals and hence they are **colourless.**

---

39. Describe the preparation of potassium dichromate from iron chromite ore.

A: Preparation of potassium dichromate ($K_2Cr_2O_7$) from chromite ore ($FeCr_2O_4$):

1) Potassium dichromate ($K_2Cr_2O_7$) is obtained by the fusion of chromite ore ($FeCr_2O_4$) with sodium (or potassium) carbonate (Na_2CO_3) in the presence of excess of air.

$4FeCr_2O_4 + 8Na_2CO_3 + 7O_2 \rightarrow 8Na_2CrO_4 + 2Fe_2O_3 + 8CO_2$.

(yellow solution)

2) The yellow solution of sodium chromate is filtered and acidified with sulphuric acid to give a solution from which orange sodium dichromate ($Na_2Cr_2O_7.2H_2O$) can be crystallised.

$2Na_2CrO_4 + 2H^+ \rightarrow Na_2Cr_2O_7 + 2Na^+ + H_2O$

3) $Na_2Cr_2O_7$ is more soluble than potassium dichromate. So the solution of sodium dichromate is treated with potassium chloride to prepare $K_2Cr_2O_7$.

$Na_2Cr_2O_7 + 2KCl \rightarrow K_2Cr_2O_7 + 2NaCl$

Orange crystals of potassium dichromate crystallise out.

[☞ Remaining SAQs are in Page 93]

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5 COORDINATION COMPOUNDS

4 Marks

40. Explain Werner's theory of coordination compounds with suitable examples.

[AP 16, 17,18,20,22,23,24,25]

A: 1) **Werner's theory** explains the structure, bonding, and isomerism of complex compounds. Here the central metal atom shows a) Primary valency b) Secondary valency.

2) Primary Valency:

i) It is equal to the oxidation number of the central atom.
ii) It is satisfied only by the negative ions.

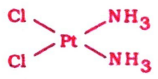
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3) Secondary Valency:

i) It is equal to the co-ordination number of the central atom.
ii) It is satisfied by negative ions, neutral molecues and rarely by positive ions.

4) Example: $[Pt(NH_3)_2Cl_2]$

i) Here, primary valency of Pt is 2.
It is satisfied by two Cl^- ions.
ii) Secondary valency of Pt is 2.
It is satisfied by two NH_3 molecules.
iii) Shape of this complex is Square.



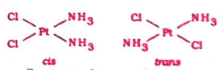
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41. Explain geometrical isomerism in coordination compounds giving suitable examples.

A: Geometrical isomerism occur due to different geometric arrangements of the ligands.

i) Square planar complexes of the type $[MX_2L_2]$:

Here **4 ligands occupy 4 corner places** of a square around the central ion.



Square planar structure

a) cis isomer: Two identical X ligands are arranged adjacent to each other.

b) trans isomer: Two identical X ligands are arranged opposite to each other.

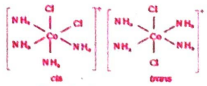
Ex: cis $[Pt(NH_3)_2Cl_2]$, trans $[Pt(NH_3)_2Cl_2]$

ii) Octahedral complexes:

Here **6 ligands occupy 6 corner places** of an octahedral.

cis & trans isomer of the type $[MX_2L_4]$:

- **cis isomer:** Two X ligands are arranged adjacent to each other.
- **trans isomer:** Two X ligands are arranged opposite to each other.



Octahedral structure

Ex: cis $[CoCl_2(NH_3)_4]^+$, trans$[CoCl_2(NH_3)_4]^+$

42. What are homoleptic and heteroleptic complexes? Give one example for each.[BMP]

A: 1) **Homoleptic complexes** are the complexes in which a metal is bound by only one kind of ligands.

2) **Ex:** $[Co(NH_3)_6]^{+31}$

3) **Heteroleptic complexes** are the complexes in which a metal is bound by more than one kind of ligands.

4) **Ex:** $[Co(NH_3)_4Cl_2]^+$

---

43. Explain the following terms by giving an example

i) Unidentate ligand **ii)Bidentate ligand**

iii) Polydentate ligand **iv)Ambidentate ligand**

A: **i) Unidentate:** It is a ligand with **one donor** atom. **Ex:** $:NH_3$

ii) Bidentate: It is a ligand with **two donor** atoms. **Ex:** $\ddot{N}H_2 - \ddot{N}H_2$

iii) Polydentate: It is a ligand with **more than two** donor atoms. **Ex:** EDTA

iv) Ambidentate ligand: It is a ligand with two donor atoms, but only one of them forms a coordinate bond at a time with central metal atom / ion.

Ex: nitro–N (or) Nitrito-N (N donor atom) $M \leftarrow N(=O)-O$

nitro-O (or) Nitrito-O (O donor atom) $M \leftarrow O-N=O$

---

44. Using IUPAC norms write the formulas for the following :

(i) Tetrahydroxozincate (II) ion **(ii) Hexaaminecobalt (III) sulphate**

(iii) Potassium tetrachloropalladate(II) **(iv) Potassium tri(oxoalato)chromate(III)**

A: i) Tetrahydroxozincate (II) ion: $[Zn(OH)_4]^{-2}$

ii) Hexaaminecobalt (III) sulphate : $[Co(NH_3)_6]_2(SO_4)_3$

iii) Potassium tetrachloropalladate(II): $K_2[PdCl_4]$

iv) Potassium trioxalatochromate(III): $K_3[Cr(C_2O_4)_3]$

---

45. Using IUPAC norms write the systematic names of the following: [AP 15,19]

(i) $[Co(NH_3)_6]Cl_3$ **(ii)** $[Pt(NH_3)_2Cl(NH_2\ CH_3)]Cl$

(iii) $[Ti(H_2O)_6]^{3+}$ **(iv)** $[NiCl_4]^{2-}$

A: i) $[Co(NH_3)_6]Cl_3$: **Hexaammine Cobalt (III)chloride**

ii) $[Pt(NH_3)_2Cl(NH_2\ CH_3)]Cl$: **Diammine chloro Methyl amine platinum(II)chloride**

iii) $[Ti(H_2O)_6]^{3+}$: **Hexaaquotitanium (III) ion**

iv) $[NiCl_4]^{2-}$: **Tetrachloronickelate (II) ion**

[☞ Remaining SAQs are in Page 93]

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6 HALOALKANES & HALOARENES

4 Marks

46. Explain the mechanism of S_N1 with example. **[BMP]**

A: **SN^1 reaction:**

1) Order of SN^1 reaction is one.

2) SN^1 is a 'unimolecular substitution reaction'.

3) SN^1 mechanism takes place in two steps.

4) **Ex:** Hydrolysis of 2-Bromo 2- methyl propane with aq KOH forms tert-butyl alcohol.

5) Mechanism of S_N1 :

Step 1: Formation of Carbocation .

$$(CH_3)_3C-Br \xrightarrow{\text{Slow step}} (CH_3)_2\overset{+}{C}CH_3 + Br^-$$

Carbocation

6) Step 2: Formation of product by the attack of Nucleophile.

$$(CH_3)_2\overset{+}{C}CH_3 + OH^- \xrightarrow{\text{Fast step}} HO-C(CH_3)_3 \quad + CH_3-C(CH_3)_2-OH$$

Carbocation — Inversion — Retention

---

47. How will you carry out the following conversions?

(i) Ethane to bromoethane **(ii) Toluene to benzyl alcohol**

A: **i) Ethane to bromoethane:** $CH_3-CH_3 + Br_{2(g)} \xrightarrow{\text{Sunlight}} Br-CH_2-CH_3 + HBr$

ii) Toluene to benzyl alcohol: When chlorine is passed into boiling toluene it gives benzyl chloride, which on boiling with KOH (aq) gives benzyl alcohol.

$$C_6H_5CH_3 \xrightarrow[-HCl]{Cl_2\Delta} C_6H_5CH_2Cl \xrightarrow{KOH(aq)} C_6H_5CH_2OH + KCl(aq)$$



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48. Describe the stereochemical effect on the hydrolysis of 2-bromobutane.

A: The given reaction is S_N2:

1) Order of S_N2 reaction is two.

2) S_N2 is a 'bimolecular substitution reaction'.

3) S_N2 mechanism takes place in single step.

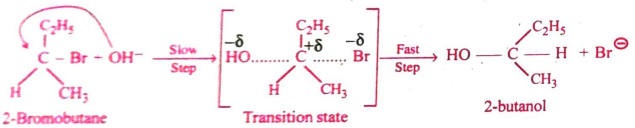
4) Given example Hydrolysis of 2-bromobutane in aqueous KOH forms **2-butanol.**

5) Reaction: C_2H_5–CH(Br)–CH_3 + KOH(aq) ⟶ C_2H_5–CH(OH)–CH_3 + KBr

2-Bromobutane → 2-butanol

6) Mechanism (S_N2 reaction): The nucleophile (OH^-) approaches the carbon from the opposite side of the halogen atom (Br) to form transition state. It is the slowest step.

7) The product has inverted configuration (Walder inversion).



49. Write the differences between S_N1 and S_N2 reactions.

A:

S_N1 reaction	S_N2 reaction
1) **Order** of S_N1 reaction is **one.**	1) **Order** of S_N2 reaction is **two.**
2) S_N1 mechanism takes place in **two steps.**	2) S_N2 mechanism takes place in **one step.**
3) S_N1 is a 'unimolecular substitution reaction'.	3) S_N2 is a 'bimolecular substitution reaction'.
4) Order of reactivity of alkyl halides : 3° > 2° > 1°	4) Order of reactivity of alkyl halides : 1° > 2° > 3°
5) Rate of reaction of S_N1 depends on concentration of reactants only.	5) Rate of reaction of S_N2 depends on concentration of reactants and nucleophile.
6) **Ex:** Hydrolysis of **2-Bromo 2-methyl propane** with aqueous KOH forms **tert-butyl alcohol.**	6) **Ex:** Hydrolysis of **2-bromobutane** in aqueous KOH forms **2-butanol.**
7) When optically active alkyl halide is subjected to S_N1 reaction, the product formed is **racemic mixture** (*d* & *l* isomers)	7) When optically active alkyl halide is subjected to S_N2 reaction, the product formed is an **inverted product.** (Walden Inversion)

50. Give the IUPAC names of the following compounds.

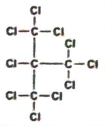
(i) CH_3–CH(Cl)–CH(I)CH_3 (ii) $ClCH_2$.CH=CH–CH_2Br

(iii) $(CCl_3)_3CCl$ (iv) $CH_3C(p\text{–}Cl\text{–}C_6H_4)_2$–CH(Br)$CH_3$

A: (i) CH_3–CH(Cl)–CH(I)CH_3 : **2-chloro, 3- iodobutane**

(ii) $ClCH_2$.CH=CH–CH_2Br: **1-bromo-4-chlorobut-2ene**

(iii) $(CCl_3)_3CCl$: **1,1,1,2,3,3,3 hepta chloro 2–(trichloromethyl)propane**



(iv) $CH_3C(p\text{–}Cl\text{–}C_6H_4)_2$–CH(Br)$CH_3$: **3–bromo-2,2-bis(4-chlorophenyl)butane**

51. Write the structures of the following organic halides.

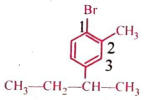
(i) 1-bromo-4-sec-butyl-2 methylbenzene

(ii) 2-chloro-1-phenylbutane

(iii) p-bromochlorobenzene

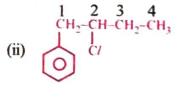
(iv) 4-t-butyl-3-iodoheptane **[AP 15]**

A: (i)



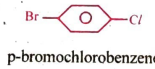
1-bromo-4-sec-butyl-2 methylbenzene

(ii)



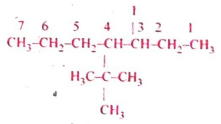
2-chloro-1-phenylbutane

(iii)



p-bromochlorobenzene

(iv)



4-t-butyl-3-iodoheptane

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[☞ Remaining SAQs are in Page 94]

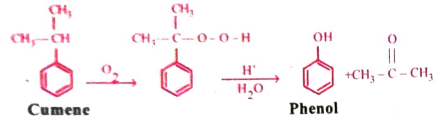
7 ALCOHOLS, PHENOLS & ETHERS

4+4 Marks

52. Give the equations for the preparation of phenol from Cumene. [BMP]

A: 1) **Cumene** is the common name for Isopropylbenzene .

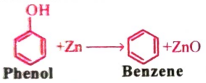
2) **Preparation:** When Cumene is oxidised in the presence of air, it is converted into cumene hydroperoxide. When treated with dilute acid, it is then converted into phenol and acetone.



Cumene **Phenol**

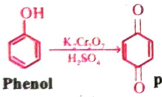
53. Write the products formed by the reduction and oxidation of phenol.

A: a) **Reduction:** Phenols on distillation with zinc dust undergo reduction to form **benzene**



Phenol **Benzene**

b) **Oxidation:** Phenol undergo oxidation with chromic acid to produce conjugated diketone which is known as p-benzoquinone.



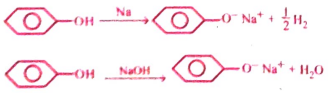
Phenol **p-benzoquinone**

54. Explain the acidic nature of phenols and compare with that of alcohols. [AP 17] [BMP]

A: I) **Acidic nature of phenols:** Phenols are weakly acidic

a) Phenols turn blue litmus to red.

b) Phenols react with sodium metal and NaOH to form H_2 and H_2O.



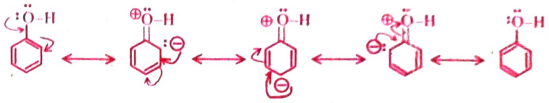
II) Acidic nature of phenols Vs alcohols:

Phenols are relatively more acidic than alcohols and also water.

a) The hydroxyl group in phenol is directly attached to SP^2 carbon on benzene ring. The carbon acts as an electron withdrawing group and Phenol exhibits resonance.

In these structures oxygen carries positive charge. This oxygen attracts electron pair of O–H bond strongly towards itself and facilitates the release of H^+ ion and $C_6H_5O^-$.

The formed phenoxide ion is more stable than phenol.



55. While separating a mixture of ortho and para nitrophenols by steam distillation, name the isomer which will be steam volatile. Give reason.

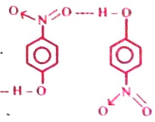
A: 1) The ortho and para isomers can be separated by steam distillation.
The ortho nitro phenol is steam volatile due to intra molecular hydrogen bonding.

2) The para-nitrophenol is less volatile due to intermolecular hydrogen bonding which results in the association of molecules.



3) o- nitrophenol shows intra molecular hydrogen bonding.

So, it is relatively more volatile than p-isomer.

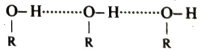


4) p-nitrophenol shows inter molecular hydrogen bonding.

So, this is less volatile than o-isomer.

56. Account for the statement: Alcohols boil at highest temperature than hydrocarbons and ethers of comparable molecular masses.

A: Alcohols have intermolecular attractions due to intermolecular hydrogen bonds. This type of intermolecular hydrogen bonding do not exist in hydrocarbons and ethers. Hence alcohols boil at higher temperature than hydrocarbons and ethers of comparable molecular masses.



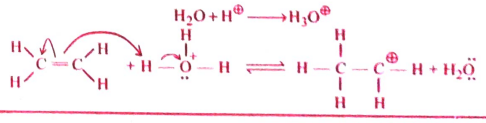
57. Write the mechanism of hydration of ethene to yield ethanol.

A: 1) Ethene is converted into alcohol by direct addition of water in the presence of phosphoric Acid (H_3PO_4).

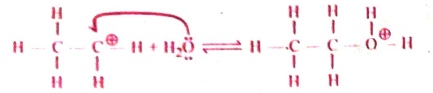
$$CH_2 = CH_2 + H_2O \xrightarrow{H^+} CH_3CH_2OH$$

2) **Mechanism(S_N1):** The mechanism of the reaction involves the following three steps:

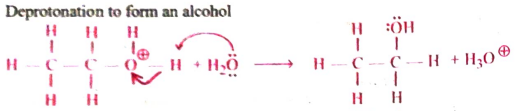
3) **Step 1:** Protonation of alkene to form carbocation by electrophilic attack of H_3O^+



Step 2: Nucleophilic attack of water on carbocation.



Step 3: Deprotonation to form an alcohol



Ethanol

---

58. Draw the structures of all isomeric alcohols of molecular formula $C_5H_{12}O$ and give their IUPAC names and classify them as primary, secondary, and tertiary alcohols.

A: The structures, IUPAC names and the classification of all isomeric alcohols of molecular formula $C_5H_{12}O$ are eight and follows as

a) $CH_3-CH_2-CH_2-CH_2-CH_2OH$	Pentan-1-ol	Primary
b) $CH_3-CH_2-CH_2-\underset{\displaystyle OH}{\underset{\vert}{CH}}-CH_3$	Pentan-2-ol	Secondary
c) $CH_3-CH_2-\underset{\displaystyle OH}{\underset{\vert}{CH}}-CH_2-CH_3$	Pentan-3-ol	Secondary
d) $CH_3-\underset{\displaystyle CH_3}{\underset{\vert}{C}}-CH_2OH$	2,2, dimethyl Propan-1-ol	Primary
e) $CH_3-CH_2-\underset{\displaystyle CH_3}{\underset{\vert}{\overset{\displaystyle CH_3}{\overset{\vert}{C}}}}-OH$	2-methylbutan-2-ol	Tertiary
f) $CH_3-\underset{\displaystyle CH_3}{\underset{\vert}{CH}}-CH_2-CH_2-OH$	3-Methylbutan-1-ol	Primary
g) $CH_3-\underset{\displaystyle CH_3}{\underset{\vert}{CH}}-\underset{\displaystyle OH}{\underset{\vert}{CH}}-CH_3$	3-Methylbutan-2-ol	Secondary
h) $CH_3-CH_2-\underset{\displaystyle CH_3}{\underset{\vert}{CH}}-CH_2-OH$	2-Methylbutan-1-ol	Primary

8 ALDEHYDES, KETONES & CARBOXYLIC ACIDS

4 Marks

59. What is Tollen's reagent? Explain its reaction with Aldehydes. **[BMP]**

A: **1) Tollen's reagent:** Ammonical silver nitrate solution is called Tollen's reagent.

Its formula is $[Ag(NH_3)_2]^+OH^-$

2) Reaction of toulene with aldehydes (Silver mirror test):

When Tollen's reagent is warmed with an aldehyde, it gets reduced to metallic silver.

This metallic silver gets deposited on the inner wall of the test tube to form a silver mirror.

$$RCHO + 2[Ag(NH_3)_2]^+ + 3OH^- \longrightarrow RCOO^- + 2Ag\downarrow + 4NH_3 + 2H_2O$$

Silver mirror

---

60. Write the oxidation products of acetaldehyde, acetone and acetophenone.

A: **1) Oxidation of acetaldehyde:** $\underset{\text{Acetaldehyde}}{CH_3CHO} \xrightarrow{(O)} \underset{\text{Acetic acid}}{CH_3COOH}$

2) Oxidation of acetone: $\underset{\text{Acetone}}{CH_3-\overset{\displaystyle O}{\overset{\Vert}{C}}-CH_3} \xrightarrow[K_2Cr_2O_7/H^+]{3(O)} \underset{\text{Acetic acid}}{CH_3COOH} + \underset{\text{Formic acid}}{HCOOH}$

3) Oxidation of acetophenone: $\underset{\text{Acetophenone}}{C_6H_5COCH_3} \xrightarrow{I_2/NaOH} \underset{\text{Idoform}}{CHI_3} + \underset{\text{Benzoicacid}}{C_6H_5COOH}$

---

61. Write the equations of any aldehyde with Fehling's reagent.

A: **1) Fehling's solution** is an alkaline solution of copper sulphate containing Rochelle salt(sodium potassium tartrate + sodium hydroxide) .

2) Fehling's test: When Fehling's solution is heated with an aldehyde, a reddish brown precipitate of (Cu_2O) is formed.

3) Reaction: $RCHO + 2Cu^{+2} + 5OH^- \longrightarrow RCOO^- + Cu_2O + 3H_2O$

62. Write the IUPAC names of the following

(i) $CH_3CH_2CH(Br)CH_2COOH$

(ii) $PhCH_2COCH_2COOH$

(iii) $CH_3CH(CH_3)CH_2COOC_2H_5$

A: i) $CH_3CH_2CH(Br)CH_2COOH$: **3-Bromo1-pentanoic acid**

ii) $PhCH_2COCH_2COOH$: **4-Phenyl-3-oxo butanoic acid**

iii) $CH_3CH(CH_3)CH_2COOC_2H_5$: **Ethyl-3-Methyl butanote**

---

63. Describe the following:

i) Crossed aldol condensation ii) Decarboxylation

A: i) Crossed aldol condensation: It is the condensation of two different carbonyl compounds (one of which must have one α-hydrogen) in the presence of alkali. It is the condensation of an aldehyde with a ketone in the presence of an alkali.

CH_3CHO + CH_3CH_2CHO → $CH_3-CH=CH-CHO$ + $CH_3-CH_2-CH=C(CH_3)-CHO$ (But-2-enal (AA); 2-MethylPent-2-enal (BB)) (Self-Aldol Products)

$CH_3-CH=C(CH_3)-CHO$ (2-Methylbut-2-enal) + $CH_3-CH_2-CH=CH-CHO$ (Pent-2-enal (BA)) (AB) (Cross Aldol Products)

Acetaldehyde (Ethanal) (A); Propionaldehyde (Propanal) (B)

ii) Decarboxylation: Sodium Ethanoate is heated with **Sodalime** to form **Methane.**

It removes a carboxyl group and releases CO_2.

Ex: $RCOONa + NaOH + CaO \xrightarrow{\Delta} R-H + Na_2CO_3$

$CH_3COONa + NaOH \xrightarrow[\Delta]{CaO} CH_4 + Na_2CO_3$

Sodium Ethanoate; Soda lime; Methane

[☞Remaining SAQs are in Page 96]

MAGICAL MEMORY MNEMONICS

- **Crossed Aldol:** Different Carbonyls + α-H
- **Decarboxylation:** DSESM [Dear SESMa మన్ను] (Remove CO_2 (One Carbon Less)

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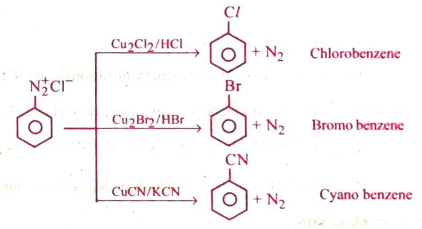
9 AMINES

4 Marks

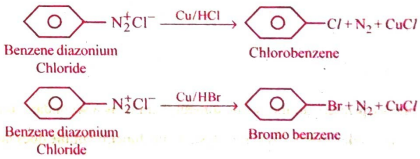
64. Explain the following name reactions: **[BMP]** [AP 15,16,17,18]

i) Sandmeyer reaction ii) Gattermann reaction

A: i) Sandmeyer reaction: Benzene diazonium chloride is treated with (Cu_2Cl_2 / HCl) (or) Cu_2Br_2 / HBr (or) CuCN/KCNto form Chloro Benzene (or) Bromo Benzene (or) Cyano Benzene.



ii) Gattermann's reaction: Benzene diazonium chloride is treated with Cu/HCl (or) Cu/HBr to form Chloro Benzene (or) Bromo Benzene



MAGICAL MEMORY MNEMONICS

- **SBdc CB-BB-CB** [State Bank delhi chowk వద్ద Chanti Babu - Bujji Babu- Chandra Babu]
- **GM Bdc CB/BB** [Guntur Mandal Brodipet Colony వద్ద China Babu- Badaa Babu]

65. Write the reactions of

(i) aromatic and (ii) aliphatic primary amines with nitrous acid, highlighting the differences.

A: i) **Aromatic** amines react with nitrous acid at low temperatures (273-278K) to form diazonium salts.

$$C_6H_5-NH_2 \xrightarrow[273-278K]{NaNO_2+2HCl} C_6H_5-N_2^+Cl^- + NaCl + 2H_2O$$

ii) Aliphatic primary amines react with nitrous acid to form aliphatic diazonium salts which being unstable, liberate nitrogen gas and alcohols. Quantitative evolution of nitrogenis used for the estimation of amino acids and proteins.

$$R-NH_2 + HNO_2 \xrightarrow{[NaNO_2+HCl]} [R-N_2^+Cl^-] \xrightarrow{H_2O} R-OH + N_2 + HCl$$

---

66. Write the equations involved in the reaction of Nitrous acid with ethylamine and aniline.

A: 1) Primary aliphatic amines like $C_2H_5NH_2$ react with nitrous acid and form aliphatic diazonium salts. But it being unstable, dissociates to give ethyl alcohol and N_2 gas.

$$C_2H_5NH_2 + HNO_2 \xrightarrow{[NaNO_2+HCl]} [C_2H_5-N_2^+Cl^-] \xrightarrow{H_2O} C_2H_5OH + N_2 + HCl$$

2) Aromatic amines like $C_6H_5NH_2$ react with nitrous acid at low temperatures (273-278K) and form **diazonium salts.**

$$\underset{\text{Aniline}}{C_6H_5NH_2} \xrightarrow[273-278K]{NaNO_2+2HCl} \underset{\text{Diazonium chloride}}{C_6H_5N_2^+Cl^-} + NaCl + 2H_2O$$

---

67. Give one chemical test to distinguish between the following pairs of compounds:

i) Methylamine and dimethylamine ii) Aniline and N-methylaniline

iii) Ethylamine and aniline

A: i) **Methylamine** is a primary amine, while dimethyl amine is a secondary amine.

So, methyl amine undergoes carbylamine test, while dimethylamine does not.

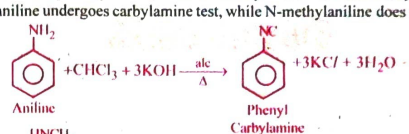
When methylamine is heated with chloroform and alcohalic KOH, an offensive smelling compound methyl carbylamine is formed.

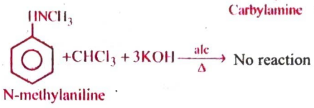
$$\underset{\text{Methyl amine}}{CH_3NH_2} + \underset{\text{Chloroform}}{CHCl_3} + 3KOH \xrightarrow[\text{heat}]{\text{alc}} \underset{\text{Methyl carbylamine}}{CH_3NC} + 3KCl + 3H_2O$$

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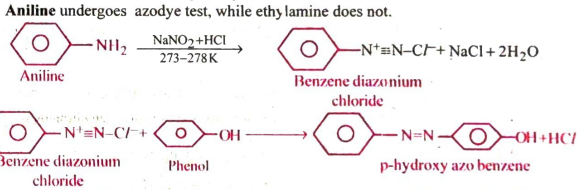
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ii) **Aniline** is a primary amine, while **N-methylaniline** is a secondary amine.
So aniline undergoes carbylamine test, while N-methylaniline does not.





iii) **Ethylamine** and aniline are primary aromatic amines.
Aniline undergoes azodye test, while ethylamine does not.



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68. Compare the basicity of the following in gaseous state and in aqueous state and arrange them in increasing order of basicity.

CH_3NH_2, $(CH_3)_2NH$, $(CH_3)_3N$ and NH_3

A: 1) pk_b values of NH_3, CH_3NH_2, $(CH_3)_2NH$, $(CH_3)_3N$ are 4.74, 3.35, 3.27, 4.22 respectively.

2) "Smaller is the value of pk_b, greater is the basic nature of amine.

Hence, **increasing order of basicity:** $NH_3 < (CH_3)_3N < CH_3NH_2 < (CH_3)_2NH$

3) In gaseous phase, basic character of amines increases with the increase in number of electron releasing groups(+I groups). Due to +I effect, the order of basicity of amines is as follows

$[(CH_3)_3N > (CH_3)_2NH > CH_3NH_2 > NH_3]$

In aqueous state, basicity depends on stability of protonated amines.

4) Greater the strength of H-bonding greater will be stability and more is the basic strength.

Hence order of basicity : $NH_3 > CH_3NH_2 > (CH_3)_2NH > (CH_3)_3N$

5) But overall basic strength of Amines depends on combined effect of steric, solvation and Inductive effect.

6) Hence overall order is **$(CH_3)_2NH > CH_3NH_2 > (CH_3)_3N > NH_3$**

10 BIOMOLECULES

4 Marks

69. Give the dietary sources of the following vitamins and name the diseases caused by their deficiency (a) A (b) D (c) E and (d) K **[BMP][AP 15,16,17,19,24]**

A:

Sl.No.	Vitamin	Sources	Deficiency diseases
1.	Vitamin A	Carrots, Fish liver oil, butter and milk	Night blindness, Xerophthalmia
2.	Vitamin D	Sunlight, fish and egg yolk	Rickets (bone deformities in children) and osteo-malacia (soft bones and joint pains in adults)
3.	Vitamin E	Vegetable oils like Sunflower oil, wheat germ oil.	Increased fragility of RBC's and muscular weakness.
4.	Vitamin K	Green leafy vegetables	Increased blood clotting time.

AC DS EV KG

---

70. What are hormones? Give one example for each. **[AP 16,18,19,23,25]**

(i) steroid hormones (ii) Polypeptide hormones and (iii) amino acid derivatives.

A: **1) Hormones :** These are the molecules which act as intracellular messengers.

They transfer biological information from one group of cells to target organs.

These are produced by Endocrine glands. They are directly released into blood.

They are basically two types (i) Animal hormones (ii) Plant hormones.

2) On the basis of their chemical nature, hormones are classified into three types.

i) Steroid hormones. Ex: Estrogens, androgens.

ii) Protein hormones. Ex: Insulin, endorphins

iii) Amino acid derivatives. Ex: Thyroxine, epinephrine, norepinephrine

Three Hormones- 4 Marks

71. Write notes on vitamins.

A: 1) Vitamins are organic compounds required in small amounts in the diet.

They are useful in the 'normal maintenance of optimum growth' and 'health of the organism'.

2) Vitamins are designated by A,B,C,D etc.,

B vitamin is further divided into subgroups. **Ex:** B_1, B_2, B_6, B_{12} etc.,

3) Vitamins are classified into two groups depending on their solubility in fat or water .

4) **Fat soluble vitamins:** These are the vitamins soluble in fat and oils but insoluble in water .

Ex: Vitamins A,D,E and K. They are stored in liver and adipose tissues.

5) **Water soluble vitamins:** These are the vitamins soluble in water.

Ex: Vitamin B, C.

They must be supplied regularly in diet because they are readily excreted in urine and cannot be stored (except vitamin B_{12}) in our body.

Additional Information (Not for Exam)

Vitamin - Sources - Deficiency diseases

Sl. No.	Name of Vitamins	Sources	Deficiency diseases
1)	Vitamin A (Retinol)	Fish liver oil, carrots, butter and milk	Night blindness, Xerophthalmia (hardening or cornea of eye)
2)	Vitamin D (Calciferol)	Exposure to Sunlight, fish and egg yolk	Rickets (bone deformities in children) and osteo-malacia (soft bones and joint pain in adults)
3)	Vitamin E (Tocopherol)	Vegetable oils like wheat germ oil, Sun-flower oil, etc.,	Increased fragility of RBC's and muscular weakness.
4)	Vitamin K	Green leafy vegetables	Increased blood clotting time.
5)	Vitamin C (Ascorbic acid)	Citrus fruits, amla and green leafy vegetables	Scurvy (bleeding gums)
6)	Vitamin B_1 (Thiamine)	Yeast, milk, green vegetables and cereals	Beri beri (loss of appetite, retarded growth)
7)	Vitamin B_2 (Riboflavin)	Milk, eggwhite, liver, kidney	Cheilosis (cracking at corners of mouth and lips), digestive disorder and burning sensation of the skin
8)	Vitamin B_6 (Pyridoxine)	Yeast, Milk, egg yolk, cereals and grams	Convulsions/Fits (rapid, involuntary muscle contraction)
9)	Vitamin B_{12}	Meat, fish, egg, curd	Pernicious anaemia (RBC deficient in haemoglobin)

72. Write the importance of carbohydrates.

A: 1) Carbohydrates are essential for the life of both plants and animals.

2) **Ex:** Honey, Starch, Cellulose.

3) Honey is an instant source of energy.

4) Starch in seen in plants and glycogen is in animals.

5) Cell walls of bacteria and plant are made up of 'cellulose'.

6) Wood and cotton fibre contain cellulose.

7) "Streptomycin" is a carbohydrate antibiotic.

8) Anticarcinogenic carbohydrates are synthesised to fight against cancer.

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73. Explain the denaturation of proteins.

A: 1) **Denaturation:** When a protein in its native form, is subjected to physical change such as change in temperature, or chemical change like change in pH, the hydrogen bonds are disturbed.

2) The protein globules unfold, helix get uncoiled and protein loses its biological activity.

3) This is called **denaturation of proteins**.

4) Secondary and tertiary structures are destroyed but the primary structure remains as such.

Ex: Curdling of milk, Coagulation of egg by boiling etc.,

---

74. What are enzymes? Give examples.

A: 1) **Enzymes** are biological catalysts. Each biological system requires a different enzyme. Enzymes are specific and efficient in their action.

2) They are required in only small quantity and work at optimum temperature (310K) and pH (7.4) under one atmospheric pressure. Chemically, they are globular proteins.

3) However some enzymes are also associated with some non-protein components called the co-factors for their activity.

4) **Ex:** Invertase, zymase, Diastase, Maltase, Urease, Pepsin



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